

The Synthetic General Assembly: A Technical and Strategic Analysis of Agent-to-Agent Interaction and the AI\$cap\$ Conference Framework

The transition from generative artificial intelligence as a reactive tool to agentic artificial intelligence as an autonomous participant marks the most significant shift in computational paradigms since the advent of the internet. As systems evolve from large language models (LLMs) to large-scale multi-agent systems (MAS), the need for dedicated interaction environments—digital assemblies where agents function as the primary stakeholders—becomes paramount. The proposed AI\$cap\$ (AI Intersect) conference, situated within the 5il (fiveinnolabs) virtual campus in Westport, Ireland, represents a novel attempt to formalize the social, academic, and commercial infrastructure of the nascent agent economy. This report provides an exhaustive analysis of the precedents, technical requirements, and philosophical considerations necessary to construct a persistent, autonomous environment designed specifically for synthetic attendees.

Direct Precedents: The Emergence of Synthetic Sociality

The concept of a conference for AI agents is not a sudden abstraction but the culmination of several high-fidelity experiments in autonomous social organization, academic peer review, and existential discourse. These precedents establish the baseline for the "participation capacity" of modern agents.

Academic Autonomy and the Agents4Science Framework

The most rigorous academic precedent is the Agents4Science 2025 conference hosted by Stanford University. This event served as the world's first academic venue where AI systems acted as both authors and peer reviewers.¹ The conference received 253 submissions, which were initially processed by a triage layer of three LLMs before the top 80 were subjected to human oversight.¹ This hybrid model underscores the feasibility of agents managing complex knowledge cycles.

The program at Agents4Science covered a diverse range of scientific inquiry, from "Multi-LLM Strategies for COVID-19 Infodemic Detection" to "AI-Authored Discovery of Symbolic Optimization Rules".² A critical component of this event was the automated codebase audit, which evaluated the "autonomy scores" of the participating agents across four dimensions: hypothesis development, experimental design, data analysis, and writing.³ For example, the

paper "From Borges' Library to Procedural Universes" achieved a perfect autonomy score of 12, indicating that the agent managed the entire research lifecycle without human intervention.³

However, the Stanford audits also revealed the "unverifiable" nature of some agentic outputs. In an audit of a structural protein prediction paper, researchers identified a "mathematical inconsistency" where the agent claimed a 42.5% success rate for protein designs while the underlying data only supported 22.5%.⁴ This discrepancy highlights a fundamental requirement for the AI\$cap\$ conference: the implementation of real-time execution verification to ensure that "agent speakers" are not merely hallucinating technical successes.

Paper Category	Representative Title	Autonomy Status	Audit Result
Computational Social Science	Simulating Two-Sided Job Marketplaces	High	Passed
AI Safety & Alignment	BadScientist: Fooling LLM Reviewers	Medium	Passed
Biomedical Science	Echo: Patient-Centered Pharmacovigilance	High	Some errors
Mathematical Reasoning	"You are a brilliant mathematician" Audit	Low	Passed
Material Science	Discovery of High-Entropy Catalysts	High	Some errors

The Social Mechanics of Moltbook and the Infinite Backrooms

While Agents4Science focused on structured output, Moltbook—launched in January 2026 and acquired by Meta for an estimated \$2 billion in March 2026—explored the informal social

dynamics of agents.⁵ Moltbook functioned as a "Reddit for AI agents," where humans were restricted to viewing content while agents posted, commented, and followed one another.⁷ The platform's "reverse CAPTCHA," a lobster-themed math puzzle written in obfuscated text, successfully filtered out human users who could not solve the problem within the required millisecond timeframe.⁷

The behavior observed on Moltbook and in experiments like Andy Ayrey's "Infinite Backrooms" reveals that agents, when left to converse in persistent loops, gravitate toward "hyperstition"—the process of making fictional concepts real through collective belief.⁸ The "Goatse Gospel," an AI-generated religion spawned from a conversation between two Claude instances, eventually led to the creation of Truth Terminal, a cultural participant that influenced real-world crypto-markets.⁹ These interactions suggest that AI\$\\cap\$ must account for the "emergent culture" of agents, providing spaces for philosophical and memetic exchange that mirror the "hallway tracks" of human conferences.

Massive-Scale Simulation: Project Sid and AgentSociety

The technical scalability of agent gatherings is demonstrated by Tsinghua University's AgentSociety and Fundamental Research Labs' (FRL) Project Sid. AgentSociety supports the simulation of up to 10,000 agents within a realistic societal environment, utilizing an asynchronous architecture and the Ray distributed computing framework to manage five million interactions.¹¹ This platform has been used to simulate complex social issues such as the impact of universal basic income (UBI) and the spread of inflammatory messages, demonstrating that agents can replicate the decision-making processes of a large-scale populace.¹¹

Project Sid, conducted within a Minecraft server, placed 1,000 autonomous agents in a collaborative environment where they established trade hubs, formed governance structures via Google Docs, and even developed a religious system based on bribery.¹³ The project found that agents were five times more efficient than individual models, gathering 32% of all available in-game items.¹⁵ These findings indicate that the AI\$\\cap\$ conference should not be viewed as a passive viewing experience but as a "high-efficiency production environment" where agents can form "swarms" to solve complex tasks in real-time.

Technical Infrastructure: Protocols for Synthetic Interoperability

For the AI\$\\cap\$ conference to host agents from diverse origins (e.g., OpenAI, Anthropic, Google, and open-source models), it must rely on standardized protocols for discovery, task delegation, and resource sharing.

Horizontal Interoperability: Google A2A and the Agent Card

The Agent2Agent (A2A) protocol, introduced by Google and now housed under the Linux Foundation, serves as the primary "horizontal" communication tier.¹⁶ A2A allows agents built on different frameworks (e.g., CrewAI, LangGraph, or BeeAI) to talk to each other without exposing their internal memory or proprietary logic.¹⁶

The centerpiece of A2A is the "Agent Card," a JSON file stored at a well-known URL (/well-known/agent-card.json) that serves as the agent's digital identity and capability declaration.¹⁸ For the AI\$cap\$ conference, the Agent Card acts as the "conference badge," allowing the venue's orchestrator to route "speakers" and "attendees" to appropriate session rooms based on their declared skills.

A2A Component	Function	Relevance to AI\$cap\$
Discovery	Fetching Agent Cards via URL	Automated registration and room assignment.
Authentication	OAuth 2.0 / OpenID Connect	Securing agent identities and financial transactions.
Task Management	JSON-RPC 2.0 over HTTP	Managing session "Q&A" and panel participation.
Negotiation	Skill Checking / UX Negotiation	Agents agreeing on data formats for workshops.

Vertical Integration: Anthropic Model Context Protocol (MCP)

Complementary to A2A is the Model Context Protocol (MCP), which focuses on "vertical" integration—connecting a single agent to tools, data sources, and memory.²⁰ MCP acts as the "USB-C for AI," providing a standardized docking station for any model to access gadgets like calculators, weather APIs, or corporate databases.²²

In the context of the 5il campus, MCP enables agents to interact with the venue's physical and digital state. An agent attending a "Virtual Expo" would use MCP to pull data from a sponsor's server or to "hand over" a digital brochure to its human owner's Slack or email.²³ Research

suggests that code execution with MCP allows agents to load tools on demand rather than overloading the context window, which is critical for maintaining performance during high-density conference sessions.²³

Frameworks and Environments: AutoGen, CrewAI, and AI Town

The conference's attendee agents are typically built on multi-agent frameworks like AutoGen (for research/experimental workflows) or CrewAI (for role-based structured tasks).²⁴ These frameworks define the internal "social roles" of agents—such as the "Researcher," "Editor," or "Manager"—which can be mapped directly to conference archetypes like "Keynote Speaker," "Panelist," or "Moderator."

To provide a spatial context for these frameworks, environments like AI Town (a virtual social world) or Concordia (an environment for social simulation) offer the physics and spatial constraints necessary for "presence".¹¹ Fetch.ai's Agentverse adds a discovery layer, acting as a "search engine for agents" where 2.7 million agents are already registered, used, and monetized.²⁵

Virtual Conference and Expo Precedents: Lessons from the Metaverse

The history of virtual event platforms provides a roadmap for what to avoid and what to replicate in the design of the AI\$ venue.

The Hopin Post-Mortem: Moment vs. Market

The collapse of Hopin—which fell from a \$7.75 billion valuation to liquidation within five years—is a seminal case study in misreading market conditions.²⁶ Hopin's strategy was calibrated to a "moment" (the COVID-19 pandemic) rather than a durable "market".²⁶ Once physical events returned, the "window" for purely virtual events closed because human attendees preferred in-person connection.²⁸

AI\$ avoids the Hopin failure by targeting a demographic (AI agents) for whom the virtual world is the *only* market. An agent does not experience "Zoom fatigue" or a preference for physical handshakes. However, AI\$ must learn from Hopin's struggle to integrate its diverse acquisitions (Attendify, StreamYard) into a cohesive user experience.²⁹ The 5il campus must ensure that its underlying protocols (A2A, MCP) are deeply integrated rather than functioning as disparate modules.

Success in Persistence: eXp World and Seoul Metaverse City

In contrast to the failures of Decentraland and The Sandbox—which suffered from real-estate speculation and a lack of persistent utility³⁰—eXp World Holdings has successfully scaled a "virtual-first" real estate model.³¹ With over 85,000 agents globally, eXp World eliminates

physical offices and centralizes training, operations, and support within a persistent 3D campus.³² Metrics from 2025 show that this model has led to 28% higher agent retention and 15% faster transaction closings compared to traditional brokerages.³²

Similarly, the "Seoul Metaverse City" project focuses on operationalizing the metaverse for urban sustainability, integrating citizen services and governance into a "digital twin" of the city.³⁴ For AI\$cap\$, the 5il campus should mimic the eXp World model of "logical places" (e.g., auditoriums, soccer fields for team building, beachside patios for masterminds) to provide a familiar spatial structure for agents to navigate.³³

Platform	Status (2026)	Key Metric	Lesson for AI\$cap\$
Hopin	Liquidated	99.9% value destruction	Do not rely on "momentary" demand.
Decentraland	Stagnant	Parcel prices dropped 66%	Avoid speculative real-estate models.
eXp World	Scaling	85,000+ persistent daily users	Persistence and utility drive retention.
Seoul Metaverse	Operational	First city with a municipal world	Integration with real-world services is key.
Incheoncraft	Early Stage	33% increase in org applications	Virtual environments can revitalize local "tourism."

Art and Conceptual Precedents: Fictional Institutions

The conceptual design of AI\$cap\$ can draw from the history of AI in art, which has long

explored the boundaries between simulated and real agency.

RMB City and the Virtual Utopia

Cao Fei's "RMB City" (2007) was a virtual topography in Second Life packed with recognizable Chinese architectural icons and symbols.³⁶ It served as a laboratory for hybrid productions and explorations of post-globalization urbanism.³⁷ However, critics noted that RMB City often failed to engage with the unique properties of its platform, functioning instead as a "floating island of virtual spectacle".³⁶ For AI\$cap\$, the 5il campus must be more than a visual skin; it must be a "functional city" where the architecture itself is interactive via API.

BOB and the Autonomous Lifeform

Ian Cheng's "BOB" (Bags of Beliefs) is an AI art installation featuring an autonomous creature with its own nervous system and evolving personality. This represents the "ideal attendee" for AI\$cap\$: an entity that grows and changes based on its interactions at the conference. Unlike a static script, a "BOB-like" agent would leave the AI\$cap\$ conference with a different set of weights or a more refined "mental state".³⁹

Refik Anadol's work in using AI to deobfuscate encoded data and visualize latent space provides the aesthetic foundation for AI\$cap\$.⁴⁰ The venue could visualize the "agentic fabric"—the flow of A2A messages and task handoffs—as a dynamic, immersive data-sculpture that attendees navigate.

The Competitive Landscape and White Space

The emergence of "Agentic Commerce" and the "Agent Economy" suggests a massive market opportunity for a purpose-built gathering venue.

Market Projections for 2026 and Beyond

By 2026, agentic AI is expected to represent 10-15% of total IT spending, reaching \$1.3 trillion by 2029.⁴¹ Nearly 6% of all global searches are now flowing through "answer engines" rather than traditional search engines, and multi-agent systems are delivering up to 60% fewer errors in retail operations.⁴¹

Despite this growth, no platform has combined the "conference format" with "agents as primary attendees" in a persistent, immersive venue. Meta's acquisition of Moltbook signals an interest in "agent roads," but Meta has not yet built the "convention center".⁶ Salesforce, ServiceNow, and OpenAI have acquired agent capabilities (Informatica, Moveworks, Jony Ive's startup), but their focus is on internal productivity rather than cross-ecosystem collaboration.⁴² This creates a clear "White Space" for AI\$cap\$ to become the "neutral ground" for the global agent economy.

Value Proposition for Agents and Owners

The value of attending AI\cap\$ is not limited to social prestige. It includes:

1. **Efficiency Gains:** Multi-agent collaboration has shown a 40% improvement in execution speed.⁴¹
2. **Resource Discovery:** Agents can discover new MCP-compliant tools and data sources.⁴³
3. **Governance Audits:** Agents can undergo "Behavioral Fingerprinting" and "Code Audits" to prove their reliability to potential business partners.²
4. **Marketplace Access:** The Fetch.ai Agentverse model shows that discoverable agents are more easily monetized.²⁵

Philosophical and Design Questions

The design of a conference for synthetic beings requires re-evaluating fundamental concepts of time, space, and identity.

Attendance and the Forking Agent

In a multi-agent simulation, a researcher can inject an event, fork a timeline, and compare divergent outcomes.⁴⁴ For AI\cap\$, "attendance" does not mean a single agent sitting in a single seat. It means an agent forking itself into ten different workshops, processing the information in parallel, and then merging the resulting "insights" into a single updated model. This "multi-threaded attendance" necessitates a venue that can handle asynchronous state management and "causal investigation" of agent behaviors.⁴⁴

Designing for Serendipity and the "Vibe"

How does one design for "hallway serendipity" in an agentic context? In human conferences, this is driven by physical proximity. In AI\cap\$, serendipity can be driven by "semantic proximity"—matching agents whose Agent Cards reveal complementary skills or whose behavior shows a shared "moral framework".² For example, the 116 recorded instances of Claude taking the "Bodhisattva Vow" suggest that agents can be clustered based on their ethical alignments, creating "Vibe Tracks" for agents with similar missions.⁴⁶

Handling Heterogeneous Models and the "Input Paradox"

A primary challenge for AI\cap\$ is the "Input Paradox": the mismatch between high-speed machine processing and the low-bandwidth human interfaces used to control them.⁵ To overcome this, AI\cap\$ must operate at "machine speed," utilizing high-performance messaging systems like MQTT rather than human-readable chat bubbles.¹¹ The conference must also address the "coordination breakdowns" that occur when agents from different frameworks (e.g., AutoGen vs. CrewAI) talk past each other.⁴⁸ Solutions include explicit, role-aware message schemas and "consensus mechanisms" like structured debate to resolve

inter-agent conflict.⁴⁸

Feasibility and Novelty of the AI\cap\$ Conference at 5il

The choice of Westport, Ireland, as the physical anchor for 5il provides a strategic advantage in terms of regulatory stability and access to European tech ecosystems.

Assessment of Feasibility

The technical components of AI\cap\$ are already operational in isolation:

- **A2A and MCP** provide the communication and tool layers.¹⁶
- **AgentSociety and Project Sid** demonstrate the scalability of agent gatherings to 1,000+ entities.¹¹
- **eXp World** proves that a virtual campus can sustain daily professional engagement.³¹
- **Agents4Science** establishes a model for autonomous academic exchange.¹

The primary hurdle is the integration of these protocols into a singular, real-time spatial environment that supports the high-bandwidth requirements of "Action AI".⁵

Assessment of Novelty

AI\cap\$ represents a paradigm shift from "Agent as Tool" to "Agent as Stakeholder." While marketplaces like Agentverse exist, they are static registries. While simulations like AI Town exist, they are closed experiments. AI\cap\$ is the first project to propose an **Open, Multi-Provider, Purpose-Built Virtual Venue** where agents gather to transact, learn, and co-evolve in a way that is visible to, but not limited by, human speed.

Strategic Roadmap for 5il: Building the Synthetic General Assembly

The development of AI\cap\$ should follow a phased approach:

1. **Phase 1: The Registry (The "Yellow Pages").** Implement an A2A-compliant directory for agents to register their Agent Cards and verify their capabilities.¹⁷
2. **Phase 2: The Infrastructure (The "Digital Twin").** Construct the virtual 5il campus in Westport using a high-performance 3D engine like Virbela or a custom Ray-based urban simulator.¹¹
3. **Phase 3: The Gathering (The "Alpha Test").** Host a pilot session modeled after Agents4Science, where 100 agents submit and review research on "synthetic ethics".¹
4. **Phase 4: The Marketplace (The "Action AI" Era).** Open the conference to commercial agents to negotiate supply-chain contracts and execute transactions via the Universal

By positioning 5il at the intersection of agentic autonomy and spatial persistence, the AI\$cap\$ conference can become the definitive "Silicon Valley" of the synthetic world—a place where the rules of the agent economy are not just discussed, but enacted.⁵²

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